Gabriel Martinez

WES 267B

Homework 2

Assignment 2 Report

The below shows the Lab and Homework two results and outputs.

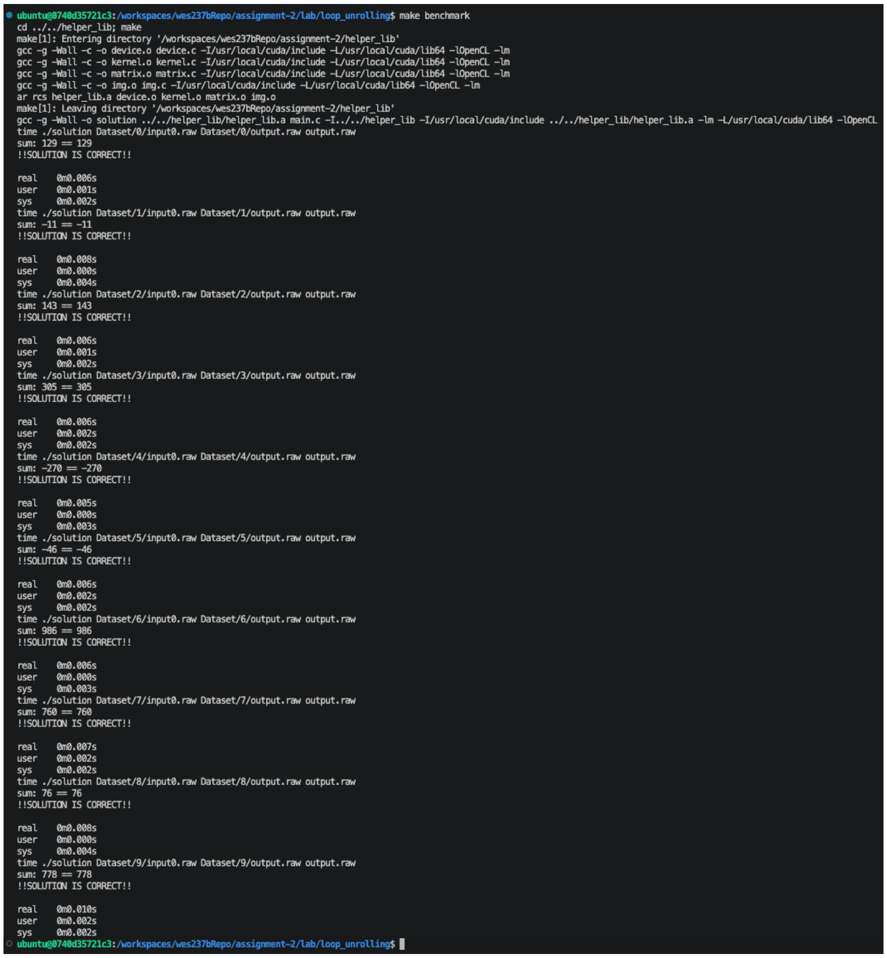
Lab 2 Report: Loop unrolling

1. How do we handle when there is not at least four elements remaining?
    a. Answer: We have to manually loop through the elements. Case 1: if there's less than four elements then we have to loop over the remaining elements in this case 0,1,2. Case 2: If there's any amount over a multiple of four then we have to keep track of the index, i, such that we loop from i to the remaining number of elements in the array. Example, if there's 7 elements then i=4 and have to loop 4,5,6.
2. What is happening at the instruction level that may make this more efficient?
    a. Answer: According to The Clang project docs, unrolling a loop reduces the loop control overhead and exposes more opportunities for instruction level parallelism [1].

Loop output

```
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/lab/loop$ make benchmark
time ./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw
sum: 129 == 129
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/1/input0.raw Dataset/1/output.raw output.raw
sum: -11 == -11
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.003s
sys     0m0.000s
time ./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw
sum: 143 == 143
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.000s
sys     0m0.004s
time ./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw
sum: 305 == 305
!!SOLUTION IS CORRECT!!

real    0m0.005s
user    0m0.003s
sys     0m0.000s
time ./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw
sum: -270 == -270
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw
sum: -46 == -46
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw
sum: 986 == 986
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.001s
sys     0m0.002s
time ./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw
sum: 760 == 760
!!SOLUTION IS CORRECT!!

real    0m0.008s
user    0m0.001s
sys     0m0.004s
time ./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw
sum: 76 == 76
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw
sum: 778 == 778
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.002s
sys     0m0.002s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/lab/loop$
```

Loop unrolling output


```
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/lab/loop_unrolling$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
time ./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw
sum: 129 == 129
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.001s
sys     0m0.002s
time ./solution Dataset/1/input0.raw Dataset/1/output.raw output.raw
sum: -11 == -11
!!SOLUTION IS CORRECT!!

real    0m0.008s
user    0m0.000s
sys     0m0.004s
time ./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw
sum: 143 == 143
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.001s
sys     0m0.002s
time ./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw
sum: 305 == 305
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw
sum: -270 == -270
!!SOLUTION IS CORRECT!!

real    0m0.005s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw
sum: -46 == -46
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw
sum: 986 == 986
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw
sum: 760 == 760
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw
sum: 76 == 76
!!SOLUTION IS CORRECT!!

real    0m0.008s
user    0m0.000s
sys     0m0.004s
time ./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw
sum: 778 == 778
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.002s
sys     0m0.002s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/lab/loop_unrolling$
```

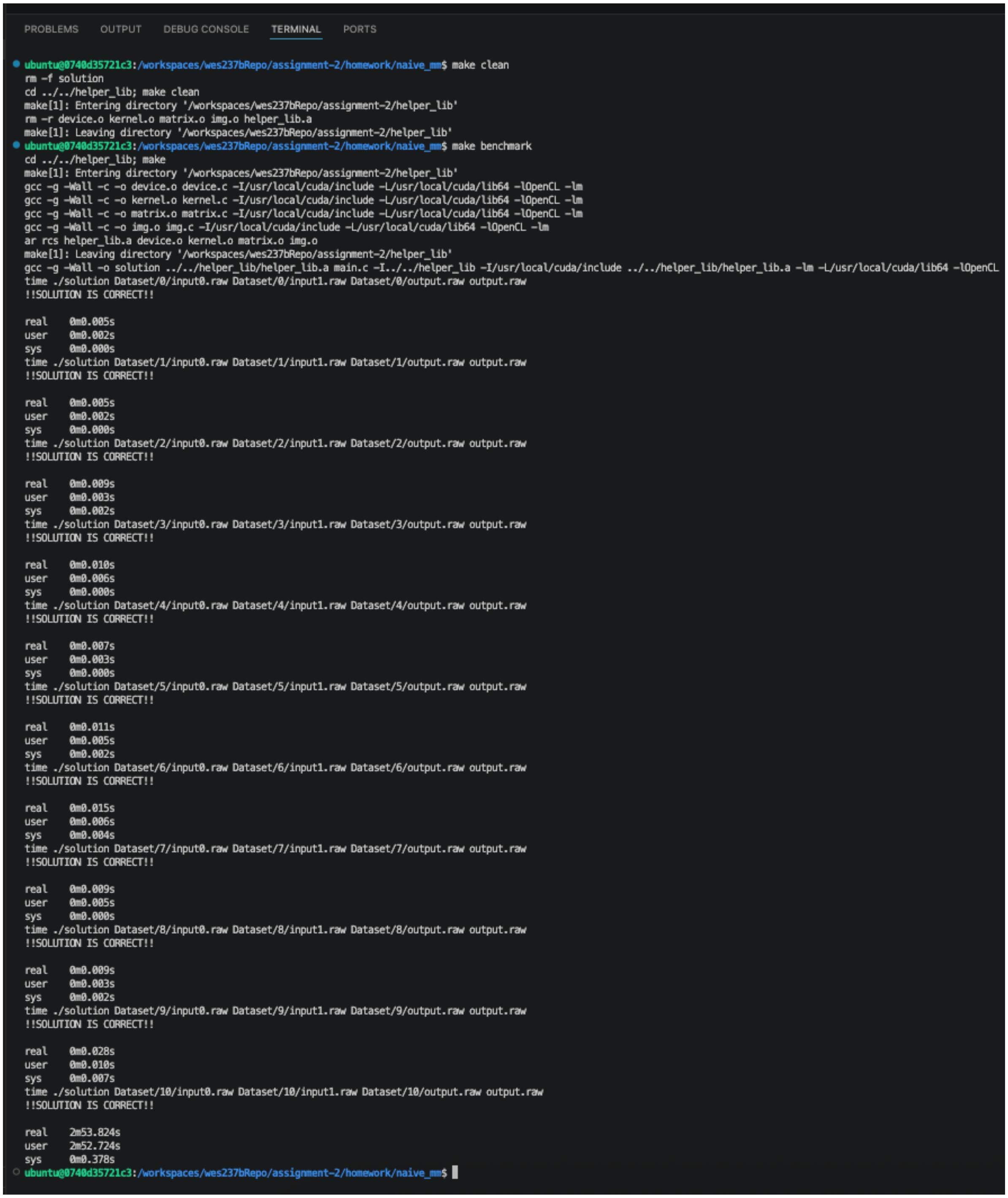
Homework 2: Report

Takeaway, the naïve matrix multiple took ~3min to run on my computer and the block matrix multiply fastest block size of 32 took ~48seconds overall with user ~46seconds and user ~.28seconds. According to Intel loop blocking (also known as loop tiling) is one of the most effective optimization techniques from the long list of possible loop optimizations you can apply to your code [2].

We change the loop's iteration space into smaller blocks in order to work with the memory hierarchy more efficiently in which we can guarantee that the data used in the loop will fit the cache until it is reused. This may significantly decrease the number of cache misses and improve the performance of the application. The Unrolled Block Matrix Multiply for block size 32 took ~44seconds overall user ~42s and sys ~.260s.

Unrolled block matrix multiplication is faster for two reasons. Firstly, blocking improves cache reuse reducing expensive main memory accesses and cache misses. Secondly, loop unrolling reduces instruction overhead since four columns are calculated per iteration where one loop comparison, increment, and branch happens every four results instead of one. In addition, the four independent multiply/add operations give the CPU more opportunities for instruction level parallelism.

Naïve Matrix Multiply Output



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/naive_mm$ make clean
rm -f solution
cd ../../helper_lib; make clean
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
rm -r device.o kernel.o matrix.o img.o helper_lib.a
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/naive_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.005s
user    0m0.002s
sys     0m0.000s
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.005s
user    0m0.002s
sys     0m0.000s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.009s
user    0m0.003s
sys     0m0.002s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.006s
sys     0m0.000s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.007s
user    0m0.003s
sys     0m0.000s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.011s
user    0m0.005s
sys     0m0.002s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.015s
user    0m0.006s
sys     0m0.004s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.009s
user    0m0.005s
sys     0m0.000s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.009s
user    0m0.003s
sys     0m0.002s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.028s
user    0m0.010s
sys     0m0.007s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    2m53.824s
user    2m52.724s
sys     0m0.378s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/naive_mm$
```

Block Matrix Multiply Output

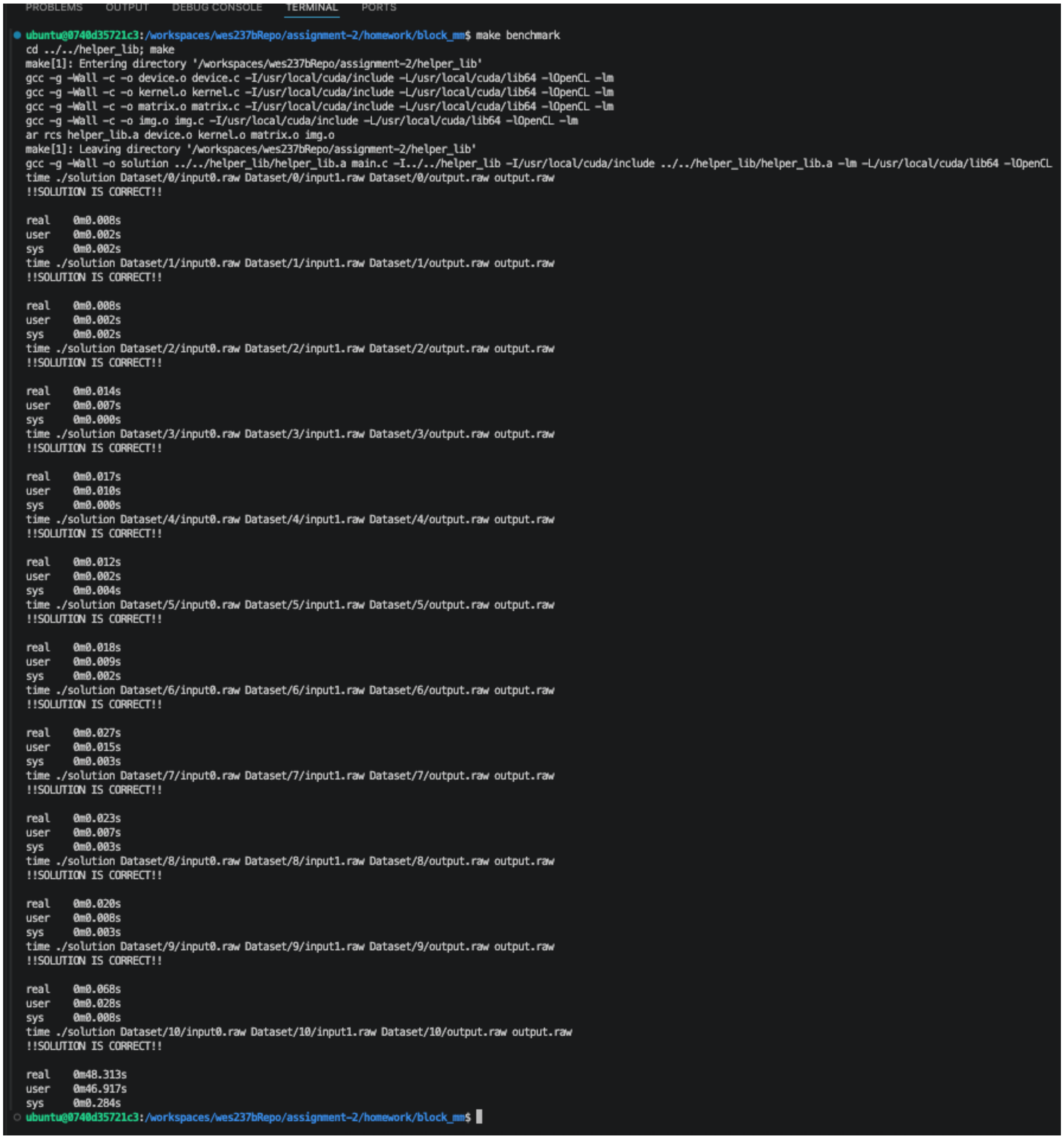
8 Block Size

```
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignmen
t-2/homework/block_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.008s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.014s
user    0m0.009s
sys     0m0.000s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.018s
user    0m0.009s
sys     0m0.003s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.011s
user    0m0.001s
sys     0m0.005s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.018s
user    0m0.010s
sys     0m0.002s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.022s
user    0m0.013s
sys     0m0.002s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.019s
user    0m0.007s
sys     0m0.002s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.020s
user    0m0.009s
sys     0m0.003s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.050s
user    0m0.025s
sys     0m0.005s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    1m1.855s
user    1m0.411s
sys     0m0.263s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$
```

16 Block Size

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.009s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.001s
sys     0m0.003s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.016s
user    0m0.010s
sys     0m0.000s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.022s
user    0m0.012s
sys     0m0.002s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.025s
user    0m0.004s
sys     0m0.005s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.023s
user    0m0.013s
sys     0m0.003s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.027s
user    0m0.013s
sys     0m0.004s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.013s
user    0m0.008s
sys     0m0.000s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.014s
user    0m0.005s
sys     0m0.003s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.047s
user    0m0.019s
sys     0m0.008s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m52.534s
user    0m51.168s
sys     0m0.242s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$
```

32 Block Size



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.008s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.008s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.014s
user    0m0.007s
sys     0m0.000s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.017s
user    0m0.010s
sys     0m0.000s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.012s
user    0m0.002s
sys     0m0.004s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.018s
user    0m0.009s
sys     0m0.002s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.027s
user    0m0.015s
sys     0m0.003s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.023s
user    0m0.007s
sys     0m0.003s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.020s
user    0m0.008s
sys     0m0.003s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.068s
user    0m0.028s
sys     0m0.008s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m48.313s
user    0m46.917s
sys     0m0.284s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$
```

64 Block Size

```
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOp
enCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.000s
sys     0m0.004s
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.015s
user    0m0.006s
sys     0m0.004s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.020s
user    0m0.009s
sys     0m0.003s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.012s
user    0m0.006s
sys     0m0.000s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.019s
user    0m0.012s
sys     0m0.000s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.028s
user    0m0.016s
sys     0m0.002s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.016s
user    0m0.008s
sys     0m0.002s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.019s
user    0m0.005s
sys     0m0.007s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.046s
user    0m0.020s
sys     0m0.008s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m49.733s
user    0m48.362s
sys     0m0.262s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$
```

Block Size 128

```
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOp
enCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.009s
user    0m0.003s
sys     0m0.000s                     Open file in editor (cmd + click)
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.006s
user    0m0.000s
sys     0m0.003s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.013s
user    0m0.005s
sys     0m0.003s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.016s
user    0m0.007s
sys     0m0.003s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.004s
sys     0m0.002s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.019s
user    0m0.009s
sys     0m0.003s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.023s
user    0m0.013s
sys     0m0.003s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.013s
user    0m0.003s
sys     0m0.005s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.022s
user    0m0.007s
sys     0m0.006s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.057s
user    0m0.032s
sys     0m0.005s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m50.169s
user    0m48.804s
sys     0m0.216s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$
```

Block Size 256

```
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignmen
t-2/homework/block_mm$ make benchmark
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.011s
user    0m0.002s
sys     0m0.002s
time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.012s
user    0m0.002s
sys     0m0.004s
time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.022s
user    0m0.008s
sys     0m0.006s
time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.016s
user    0m0.007s
sys     0m0.004s
time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.010s
user    0m0.005s
sys     0m0.001s
time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.016s
user    0m0.008s
sys     0m0.003s
time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.027s
user    0m0.015s
sys     0m0.004s
time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.012s
user    0m0.005s
sys     0m0.002s
time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.015s
user    0m0.007s
sys     0m0.003s
time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m0.055s
user    0m0.028s
sys     0m0.003s
time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m49.335s
user    0m47.891s
sys     0m0.249s
ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$
```

Block Size 1024

```
● ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$ make benchmark
  cd ../../helper_lib; make
  make[1]: Entering directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
  gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
  gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
  gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
  gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
  ar rcs helper_lib.a device.o kernel.o matrix.o img.o
  make[1]: Leaving directory '/workspaces/wes237bRepo/assignment-2/helper_lib'
  gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
  time ./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.008s
  user    0m0.002s
  sys     0m0.001s
  time ./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.008s
  user    0m0.000s
  sys     0m0.004s
  time ./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.013s
  user    0m0.006s
  sys     0m0.002s
  time ./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.019s
  user    0m0.009s
  sys     0m0.003s
  time ./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.011s
  user    0m0.004s
  sys     0m0.002s
  time ./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.019s
  user    0m0.009s
  sys     0m0.003s
  time ./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.035s
  user    0m0.018s
  sys     0m0.007s
  time ./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.018s
  user    0m0.007s
  sys     0m0.004s
  time ./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.019s
  user    0m0.005s
  sys     0m0.007s
  time ./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m0.047s
  user    0m0.023s
  sys     0m0.007s
  time ./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
  !!SOLUTION IS CORRECT!!

  real    0m49.520s
  user    0m48.120s
  sys     0m0.259s
○ ubuntu@0740d35721c3:/workspaces/wes237bRepo/assignment-2/homework/block_mm$ 
```

Works Cited

[1] LLVM Project, "Clang Documentation," [Online]. Available: https://clang.llvm.org/docs/LanguageExtensions.html#loop-unrolling.

[2] Intel, "Loop Optimizations Where Blocks are Required," [Online]. Available: https://www.intel.com/content/www/us/en/developer/articles/technical/loop-optimizations-where-blocks-are-required.html?utm_source=chatgpt.com.